A cubic weakly reversible mass-action system with an algebraic limit cycle

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17-reaction network: Claude Opus 5.5 exploration agent (effort high), 17 Oct 2026 · Rigorous certificate, 10-reaction network and manuscript: Claude Opus 5.5 (effort xhigh) · Independent verification: Claude Opus 5.5 (effort xhigh), 18 Oct 2026

Abstract. Craciun and Erban asked whether a cubic weakly reversible two-species mass-action system can have an algebraic limit cycle, i.e. whether $W^a(3) \geq 1$. We answer positively with a weakly reversible network of ten reactions, all complexes of molecularity at most three, and integer rate constants. The circle $(x-2)^2+(y-2)^2 = 1$ is invariant, carries no equilibrium, and has characteristic exponent $\oint \operatorname{div} F\, dt \in [-3.68475192932676 \pm 3.3 \times 10^{-15}]$, certified by rigorous ball-arithmetic integration (and confirmed independently by interval arithmetic and by a residue computation); it is therefore a hyperbolic, stable, algebraic limit cycle.

The question

Craciun and Erban [1] introduce chemical analogues of Hilbert's 16th problem. For $n \in \mathbb{N}$, W_n is the set of two-species mass-action systems whose reactions have at most n reactant and at most n product molecules and whose reaction graph is weakly reversible Definitions 2–3 [1]; $W^a(n)$ is the maximum number of algebraic limit cycles of systems in W_n Definition 5 [1]. Their Table 2 records $W^a(2) = 0$, $W^a(3) \geq 0$ and $W^a(4) \geq 1$, and they write (arXiv:2406.05057v2, §4 and §6):

> *"The question of whether cubic weakly reversible two-species systems can give rise to algebraic limit cycles remains open, leaving us with inequality $W^{(a)}(3) \geq 0$ in Table 2." … "it remains an open question whether a cubic weakly reversible system can have an algebraic limit cycle."*

Theorem 1. $W^a(3) \geq 1$. Explicitly, the weakly reversible network with integer rate constants

$$0 \xrightarrow{434} 2Y, \qquad 0 \xrightarrow{448} X+Y, \qquad Y \xrightarrow{3374} 0, \qquad 2Y \xrightarrow{1728} 3Y,$$
$$3Y \xrightarrow{194} Y, \qquad 3Y \xrightarrow{11} 2X+Y, \qquad X+Y \xrightarrow{328} X+2Y,$$
$$X+2Y \xrightarrow{250} 3Y, \qquad 2X+Y \xrightarrow{216} 3X, \qquad 3X \xrightarrow{28} X+2Y$$

belongs to W_3, and its mass-action system

$$\dot{x} = f(x,y) = -56x^3 + 216x^2y - 250xy^2 + 22y^3 + 448,$$
$$\dot{y} = g(x,y) = 56x^3 - 216x^2y + 250xy^2 + 328xy - 410y^3 + 1728y^2 - 3374y + 1316$$

has the circle $\mathcal{C} = \{(x-2)^2 + (y-2)^2 = 1\} \subset \mathbb{R}^2_{>0}$ as a hyperbolic, orbitally asymptotically stable limit cycle.

1 Proof

Proof (of Theorem 1). *(1) The network is in W_3.* Its eight complexes $0, Y, 2Y, 3Y, X+Y, X+2Y, 2X+Y, 3X$ have molecularity at most 3, so every reaction has at most three reactant and at most three product molecules (condition (2.10) of [1] with $n=3$). The standing assumptions on networks (2.1) of [1] also hold: no reaction is repeated, no reaction has equal source and product, and both species change in some reaction. Every reaction lies on one of the three directed cycles $0 \to 2Y \to 3Y \to Y \to 0$, $0 \to X+Y \to X+2Y \to 3Y \to Y \to 0$ and $3Y \to 2X+Y \to 3X \to X+2Y \to 3Y$, which together contain all ten reactions: the network is weakly reversible, with a single (strongly connected) linkage class. The equations follow from $\dot{z} = \sum \kappa\,(\text{product} - \text{source})\,x^a y^b$ over the reactions.

(2) The circle is invariant. With $h(x,y) = (x-2)^2 + (y-2)^2 - 1$, direct expansion gives the polynomial identity

$$f\,h_x + g\,h_y = s\,h, \qquad s(x,y) = -112x^2 + 544xy - 448x - 820y^2 + 1728y - 1008.$$

So $\dot{h} = s\,h$ along solutions and $\mathcal{C} = \{h = 0\}$ is invariant. It lies in $[1,3]^2$, inside the open positive quadrant.

(3) There is no equilibrium on $\mathcal{C}$. Parametrise $\mathcal{C}$ by $x = 2+\cos\theta$, $y = 2+\sin\theta$. The field is tangent to $\mathcal{C}$, so $F = (f,g) = P\,(-\sin\theta, \cos\theta)$ with $P = -(y-2)f + (x-2)g$, and $\dot{\theta} = P$. The resultant $\mathrm{Res}_y(P,h) = 75272x^4 - 474912x^3 + 909300x^2 - 295792x + 56532$ (the leading coefficient of h in y is 1) has no real root in $[1,3]$ (Sturm count), so P and h have no common real zero; as $P(3,2) = 792 > 0$ and $\mathcal{C}$ is connected, $P > 0$ on $\mathcal{C}$. Equivalently and more explicitly, $P \equiv -2(97x^2 - 97xy - 262x - 38y + 175) \pmod{h}$, and with $t = \tan(\theta/2)$, $P = 4(130t^4 + 135t^3 + 522t^2 + 329t + 198)/(1+t^2)^2$, whose numerator has no real root (Sturm), while $P(1,2) = 520$ at $t = \infty$; the same method applied to $P - 18759/50$ gives the exact bound $P > 375.18$ on $\mathcal{C}$. Independently, a ball-arithmetic evaluation on 4096 subintervals of $[0, 2\pi]$ gives $P \geq 348$ on $\mathcal{C}$. Consequently $\mathcal{C}$ is a periodic orbit, traversed counter-clockwise, of period $T = \int_0^{2\pi} d\theta / P(\theta) \in [0.0103320434082921 \pm 4.3 \times 10^{-17}]$.

(4) The periodic orbit is hyperbolic and stable. For a periodic orbit γ of a planar C^1 vector field, the derivative of the Poincaré return map at γ equals $\exp\left(\oint_\gamma \operatorname{div} F\,dt\right)$ §3.4, Theorem 2 [2]. Here

$$\oint_{\mathcal{C}} \operatorname{div} F\,dt = \int_0^{2\pi} \frac{\operatorname{div} F(2+\cos\theta, 2+\sin\theta)}{P(\theta)}\,d\theta \in [-3.68475192932676 \pm 3.3 \times 10^{-15}],$$

computed by rigorous ball-arithmetic integration (Arb's `acb_calc_integrate` [3], via python-flint), which returns a guaranteed enclosure. The integrand is a rational function of $\cos\theta, \sin\theta$, analytic on a neighbourhood of $[0, 2\pi]$ because P has no zero there (step (3)); its complex poles need no special treatment, since on any complex ball where P may vanish the ball division returns an indeterminate value and the integrator subdivides. Two further independent evaluations agree with this enclosure: a composite midpoint rule in mpmath interval arithmetic, with the remainder $\frac{h^3}{24}\,\phi''$ enclosed on each cell, gives $[-3.6847574, -3.6847469]$; and, with $t = \tan(\theta/2)$, the integral becomes that of a rational function over $\mathbb{R}$ with three simple poles in the upper half-plane ($t = i$ and two roots of $130t^4 + \cdots + 198$), and $2\pi i$ times the sum of its residues there is $-3.684751929326756730\ldots$ The integral is negative, so the return map is a contraction

at γ: $\mathcal{C}$ is a hyperbolic, orbitally asymptotically stable periodic orbit, in particular isolated among periodic orbits, i.e. a limit cycle. It is algebraic, being a component of $\{h = 0\}$. As a consistency check, $\oint_{\mathcal{C}} s\,dt$ has the same enclosure, as it must: the coordinate h is transversal to $\mathcal{C}$ and satisfies $\dot{h} = sh$. □

2 Remarks

Remark 2 (construction)**.** With h fixed, the conditions "$fh_x + gh_y = sh$" and "the realisation uses only complexes of molecularity at most 3" are linear in the rate constants and in the coefficients of s; weak reversibility is enforced by a strictly positive circulation on the support of the rate vector. A linear program found a 17-reaction network during exploration; a mixed-integer program minimising the number of reactions (SCIP) then returned the 10-reaction support of Theorem 1, on which the solution of the linear identity is unique up to a time rescaling. This yields the integer rate constants. (Independent floating-point check by the verifier, `verification/milp_min_support.py` with HiGHS: no network in W_3 with at most nine reactions, all rate constants in $[1, 10^5]$, $\mathcal{C}$ invariant and $|P| \geq 1$ with constant sign at 96 equally spaced points of $\mathcal{C}$ exists; ten reactions are attained for both orientations, and the optimal support is not unique. This bounded, sampled minimality is not used in the proof.) ⌟

Remark 3 (a second example)**.** The 17-reaction network N17 (10 complexes, rational rate constants), whose reactions are listed in `scripts/certify_cycle.py`, also has $\mathcal{C}$ as a stable hyperbolic limit cycle, with $\oint \operatorname{div} F\,dt \in [-0.636539668381478 \pm 2 \times 10^{-16}]$ and period $1.91598\ldots$ ⌟

Remark 4. Since $W_3 \subseteq M_3$ and $W_3 \subseteq W_4$, the example also re-proves $M^a(3) \geq 1$ and $W^a(4) \geq 1$ with a cubic system. The network has eight complexes, one linkage class and rank two, hence deficiency five; it is not covered by the deficiency-zero or deficiency-one theorems [4]. ⌟

3 Computer verification

`scripts/certify_cycle.py` checks, for both networks, in exact rational arithmetic (sympy): membership in W_3 (molecularities, strong connectivity), the invariance identity and its cofactor, the resultant and the sign of P; and in rigorous ball arithmetic (python-flint/Arb): the positivity of P on 4096 subintervals and enclosures of the period, of $\oint s\,dt$ and of $\oint \operatorname{div} F\,dt$. Runtime about 5 s.

`verification/verify_independent.py`, written by the independent verifier without access to the solver's code, re-derives the same facts by different methods: reaction-by-reaction weak reversibility and all assumptions of (2.1)–(2.10) of [1]; invariance by division and by substitution of the rational parametrisation of $\mathcal{C}$; positivity of P by a Sturm count on the tangent half-angle polynomial; rigorous enclosures of $\oint \operatorname{div} F\,dt$ and of the period with mpmath interval arithmetic (composite midpoint rule with enclosed remainder) and a residue evaluation of both integrals; a floating-point check that the monodromy determinant equals $e^{\oint \operatorname{div} F\,dt} \approx 0.0251034$; and the uniqueness, up to scaling, of the rate vector on the 10-reaction support. Runtime about 3 min.

References

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